

ABDULLA AL HUSSAIN

Software Engineer | Full-Stack Developer | AI Researcher

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PERSONAL INFORMATION

Full Name	Abdulla Al Hussain
Date of Birth	02 December 2003
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PROFESSIONAL SUMMARY

Results-driven B.Sc. CSE graduate (CGPA 3.78/4.00, Varendra University, 2026) with hands-on expertise in full-stack web development, machine learning, and applied AI research. Proficient in building scalable, production-ready applications using React, Node.js, MongoDB, and Firebase. IEEE-published researcher specialising in deep learning for medical image analysis. Demonstrated capacity to deliver end-to-end software solutions, collaborate across teams, and communicate complex technical concepts with clarity. Seeking a challenging software engineering, web development, or AI/ML role in a European academic or professional environment.

WORK EXPERIENCE

Research Assistant – Financial Data Analysis

Collaborative Research Project | Massey University MBA Research | Remote | 2026

- Conducted end-to-end data analysis investigating the causal impact of inflation on stock market returns using advanced econometric methods: Propensity Score Matching (PSM), Inverse Probability of Treatment Weighting (IPTW), and Doubly Robust (DR) estimation.
- Engineered Python data pipelines to preprocess, clean, and model large financial datasets; produced findings confirming a statistically significant negative relationship between inflation and stock returns.
- Authored comprehensive technical documentation and research reports, demonstrating strong written communication and analytical rigour.

EDUCATION AND TRAINING

B.Sc. in Computer Science and Engineering

Varendra University (UGC-Approved) | Rajshahi, Bangladesh | 2022 – 2026

- CGPA:** 3.78 / 4.00
- Key Coursework:** Data Structures & Algorithms, Machine Learning, Database Management Systems, Software Engineering, Computer Networks, Object-Oriented Programming, Operating Systems

Higher Secondary Certificate (HSC) – Science

Shahid Buddhijibi Government College | Rajshahi, Bangladesh | 2020

- GPA:** 5.00 / 5.00 — Perfect Score

Secondary School Certificate (SSC) – Science

River View Collectorate School | Rajshahi, Bangladesh | 2018

- GPA:** 5.00 / 5.00 — Perfect Score

RESEARCH AND PUBLICATIONS

IEEE ICCIT 2025 – Conference Paper

IEEE Xplore | DOI: 10.1109/ICCIT68739.2025.11491822

- Designed and implemented a deep transfer learning pipeline for the detection and multi-class severity classification of diabetic retinopathy from retinal fundus images.
- Fine-tuned pre-trained CNN architectures via transfer learning to achieve high classification accuracy on medical imaging datasets, contributing towards accessible AI-assisted ophthalmology diagnostics.
- Managed the complete research lifecycle: literature review, model development, training, evaluation, and peer-reviewed publication in IEEE Xplore.

PROJECTS

Chiisai Trees – Bonsai Collection Management System | React.js · Vite · Tailwind CSS · DaisyUI · Firebase Authentication · Google Apps Script · Google Sheets · JavaScript · [Vercel](#) | 2026

- Developed and deployed a full-stack web application to manage bonsai collections, digitising repot history, collection dates, styling records, and annual progress photos.
- Implemented secure Firebase Google Authentication with a protected admin dashboard and full CRUD functionality for bonsai management.
- Integrated Google Sheets with Google Apps Script as a lightweight serverless backend API for data storage and retrieval.
- Built using React, Vite, Tailwind CSS, DaisyUI, and Firebase, and deployed the application on Vercel.

Front-End Projects Portfolio | HTML5 · CSS3 · Tailwind CSS · DaisyUI | 2026

- Designed and published multiple responsive front-end projects on GitHub, showcasing layout design, component styling, and UI best practices.

Notepad – Real-Time Note Management App | React · Node.js · MongoDB | 2025

- Built a full-stack CRUD application enabling users to create, edit, organise, and search notes in real time, with persistent cloud storage and instant UI updates.

Hope's Rising – Full-Stack Donation Platform | React · Node.js · Firebase · REST API | 2024

- Architected and deployed a real-time, multi-category donation platform featuring secure financial transaction handling, Firebase authentication, and role-based access control.
- Implemented a responsive UI with real-time data synchronisation via Firebase Firestore, reducing data latency and supporting concurrent users across multiple donation categories.
- Integrated secure payment flows and user verification mechanisms to ensure data integrity and platform trustworthiness.

Book Store – Full-Stack E-Commerce Application | React · Node.js · MongoDB · Express.js | 2024

- Developed a complete e-commerce web application with product listing, dynamic category filtering, cart management, and secure JWT-based user authentication.
- Designed RESTful API endpoints and MongoDB schemas for efficient data retrieval, enabling scalable product and order management.

Hydro Harmony – Hult Prize Innovation Project | Global Social Entrepreneurship Competition | 2024

- Led the technical development of a safe water accessibility solution for the Hult Prize global competition; presented a scalable model to an international panel of evaluators.
- Demonstrated product thinking, cross-functional collaboration, and the ability to translate social impact goals into a technically feasible prototype.

TECHNICAL SKILLS

Programming Languages	Python, C, C++, Java, JavaScript (ES6+)
Frontend	React.js, Vite, HTML5, CSS3, Tailwind CSS, DaisyUI, Responsive Design
Backend	Node.js, Express.js, REST API Design, Google Apps Script (Serverless)
Databases	MongoDB, Firebase Firestore, Firebase Realtime Database, Google Sheets (API-backed)

AI / Machine Learning	Deep Learning, Transfer Learning, CNNs, TensorFlow, Keras, Scikit-learn
Data & Statistics	Pandas, NumPy, Econometric Modelling (PSM, IPTW, DR Estimation)
Deployment & DevOps	Vercel, Git, GitHub, CI/CD Basics
Development Tools	VS Code, Postman, npm, Firebase Console
Other	MS Office Suite (Word, Excel, PowerPoint), Technical Writing, Agile Basics

DIGITAL SKILLS

- Full-Stack Web Development: React.js, Vite, Node.js, Express.js, MongoDB, Firebase, Google Apps Script
- AI & Machine Learning: TensorFlow, Keras, Scikit-learn, CNN Architectures, Transfer Learning
- Cloud & Deployment: Vercel, Firebase (Auth, Firestore, Realtime DB), Google Sheets API
- Version Control & Collaboration: Git, GitHub
- API Development & Testing: RESTful API Design, Postman, Serverless Functions
- Data Analysis: Python (Pandas, NumPy), Econometric Modelling
- Productivity Tools: MS Word, MS Excel, MS PowerPoint, VS Code

LANGUAGE SKILLS

English-Fluent Bengali-Native Spanish-Beginner

AWARDS AND ACHIEVEMENTS

- IEEE Xplore Publication (ICCIT 2025) — Peer-reviewed international conference paper on deep learning for medical image analysis.
- Hult Prize 2024 Participant — One of the world's largest student social entrepreneurship competitions; represented Varendra University.
- HSC Perfect GPA (5.00/5.00) — Shahid Buddhijibi Government College, 2020.
- SSC Perfect GPA (5.00/5.00) — River View Collectorate School, 2018.
- Runners-up, Intra-Department Football Tournament — demonstrating leadership, teamwork, and performance under pressure.

COMMUNICATION SKILLS

- Strong technical writing ability demonstrated through a peer-reviewed IEEE publication, research reports, and professional documentation.
- Delivered a project presentation to an international panel of evaluators at the Hult Prize 2024 global competition.
- Fluent in English for professional, academic, and cross-cultural communication.

LEADERSHIP AND ORGANISATIONAL SKILLS

- Led the technical development and international presentation of the Hydro Harmony project at the Hult Prize 2024 competition.
- Managed the complete lifecycle of IEEE-published research from literature review and model development through to peer review and publication.
- Independently conceived, designed, and delivered the Chiisai Trees project — from requirements gathering to production deployment — demonstrating initiative and full product ownership.

CORE COMPETENCIES

Analytical Thinking · Problem-Solving · Time Management · Team Collaboration · Presentation Skills · Networking · Attention to Detail · Self-Motivated · Fast Learner

REFERENCES

References available upon request.